6.S985 Multimodal AI Project Proposal

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Abstract

Real estate decisions, such as search, valuation, development, underwriting, and portfolio management, rely on heterogeneous data across language, visual, and geospatial data modalities. Yet, existing approaches limit predictive performance by encoding these modalities in isolation or relying on derived secondary features, resulting in the loss of critical structural and spatial information. This proposal aims to explore how joint multimodal representations improve predictive accuracy, decision quality, and market transparency in three directions: (1) property valuation and pricing, (2) preference-conditioned property retrieval and ranking, and (3) optimal urban planning and site selection.

1 Introduction

Real Estate is fundamentally an information-fusion domain. Consequential decisions across an asset's lifecycle: search, valuation, underwriting, development, and portfolio management, require joint reasoning over heterogeneous evidence: tabular attributes (price, size, history, rents), unstructured language (listings, appraisals, zoning text), visual evidence (interior/exterior images, floor plans, satellite imagery), and geospatial context (transit access, land use, neighborhood change, environmental risk). Despite this richness, current analytics pipelines either process these signals in isolation or collapse them into coarse summaries, limiting both predictive performance and decision quality.

Real estate concentrates core multi-modal learning challenges into high-stakes settings: heterogeneous representations, imperfect cross-modal alignment, modality conflict, and selective disclosure. Persuasive listing language may overstate quality, curated photos may mask weak location fundamentals, attractive tabular metrics may coexist with hidden regulatory constraints. A robust model must fuse these signals and determine which to trust. This yields our central research problem: how can multi-modal AI systems integrate heterogeneous real estate signals to improve decision quality while remaining robust and interpretable under missing data, modality conflict, and spatial-temporal shift?

The potential impact is significant. Real Estate markets shape household wealth, business location, urban form, and municipal finance. Better decision-support systems could reduce information asymmetry and expand access to higher-quality analytics beyond large institutions. However, the risks are non-trivial, because poorly designed systems may amplify historical bias, so our research value lies not only in prediction accuracy, but also in reliability diagnostics: identifying when a modality helps, when it hurts, and when its effects inordinately dominate.

2 Related Works

The landscape of multimodal AI in real estate has evolved from simple regression models to sophisticated architectures that integrate heterogeneous data. We categorize prior work into three main themes: (i) Multimodal Real Estate Valuation, (ii) Natural Language Preference Grounding, and (iii) Optimal Urban Planning.

2.1 Multimodal Property Valuation

Current real estate tools focus on property valuation, using multimodal machine learning to combine structured tabular data (e.g., size, bedrooms, prior price) with property images to estimate sale prices.

For example, Poursaeed & Belongie (2018) [11] proposes assigning luxury levels to rooms based on interior and exterior images. For each property, room-level predictions are aggregated to form a visual feature vector and used in a regression model to predict prices. Since this method uses readily available modalities, the representation captures only a biased snapshot of a property. Additionally, reducing visual data to secondary features discards fine-grained spatial details.

Karanikolas et al. (2025) [6] expanded this by incorporating unstructured language data from property advertisements and neighborhood data, encoding text with multilingual transformers, visual data with ResNet-18, and geospatial features representing location and nearby points of interest (POIs). Each modality is separately inputted into tree-based regressors. While this methodology leverages multiple data sources, it encodes each modality independently and performs fusion only at the final stage, which can limit the model's ability to capture cross-modal interactions. In contrast, our approach explores earlier fusion strategies and richer representations, aiming to retain more detailed information from each modality and improve predictive accuracy.

2.2 Preference Grounding and Tradeoff-aware Retrieval from Natural Language Queries

Research in real estate search lies at the intersection of information retrieval, recommender systems, and multimodal learning. Traditional search relies on structured filters (price, beds, baths) that fail to capture latent preferences like "natural light" or "quiet street." Wang et al.(2021) [16] address preference ambiguity through a Fuzzy Analytic Hierarchy Process, modeling uncertainty in buyer requirements via triangular fuzzy numbers, but rely on static questionnaire data rather than rich multimodal listing signals. Platforms like Zillow now support natural language queries, but LLM-based constraint extraction tends to flatten nuanced requests into rigid filters, losing the soft tradeoff structure inherent in real housing decisions [2].

Embedding-based methods have shifted toward latent space representations. SeoulHouse2Vec (Jun et al. 2020) [5] uses dense collaborative filtering to map users and properties into shared vector spaces, surfacing recommendations that rigid filters would miss. Models like MM-Embed [9] extend this with joint text-image embeddings via CLIP [12], enabling cross-modal similarity. However, these approaches collapse signals into a single vector, making modality-specific evidence difficult to surface. For example, verifying whether a kitchen is truly "bright" by cross-referencing floor plans and geospatial orientation. Likewise, Roomsemble (Kottmyer 2022) [7] recommends properties based on a user-provided photo of their preferred aesthetic, but operates purely as image retrieval without natural language reasoning or geospatial grounding.

2.3 Optimal Urban Planning

GIS-based multi-criteria evaluation (MCE) combined with machine learning has been applied to large-scale land use suitability analysis. Radocaj et al. (2022) [13] show that Random Forest and XGBoost substantially outperform conventional GIS-MCE for cropland suitability using satellite imagery. He et al. (2024) [3] achieve 90-93% accuracy integrating ML with GIS for regional land use planning in China. However, these methods operate at coarse regional scales, focusing on environmental suitability (soil type, slope, flood risk) rather than parcel-level contextual appropriateness. They predict what can be built given physical constraints, not what should be built given contextual fit.

On the vision side, Mushkani et al. (2025) [10] benchmark VLMs on urban scene understanding, finding strong performance on objective properties (vegetation, spatial layout) but weaker performance on subjective attributes (cultural character, overall impression). Ro et al.(2025)[15] demonstrate that VLMs like LLaVA can be fine-tuned for spatial reasoning in street views using synthetic QA datasets. These works describe existing scenes but do not prescribe future development. Commercial platforms such as Deepblocks [1] use AI for development feasibility studies, but focus on financial analysis for pre-specified uses rather than recommending optimal uses from scratch.

3 Proposed contribution

3.1 Multimodal Real Estate Valuation

Prior machine learning-based real estate valuation models rely on tabular features, images, and geospatial data. However, modern real estate platforms like Zillow are increasingly including virtual 3D home tours, historical price trajectories, nearby comparable properties, and floor plans. These data sources are substantially harder to obtain and integrate as publicly available datasets, yet they encode critical information about spatial layout and functional livability. Our proposed project extends these models through incorporating these high-fidelity visual and temporal data, and evaluating how different fusion strategies affect predictive performance.

To obtain our data, we plan to scrape Zillow for sold houses in the Greater Boston area. For each listing, we will gather the following high-fidelity modalities: (1) walkthrough videos and 3D interior representations; (2) floor plans, interior/exterior photos; (3) historical price trajectories and information about nearby comparable properties as numerical data. All data will be cleaned, aligned with property identifiers, and labeled with the final sale price. Success will be evaluated by how closely predictions match final sale prices, using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

3.1.1 Planned Execution

We aim to ask the following questions: (1) Does incorporating high-fidelity, multimodal property data improve price prediction accuracy compared to traditional tabular or image-only models? (2) Which modalities contribute most to predictive performance? We hypothesize that incorporating higher-fidelity data into our models via early fusion will outperform models using only conventional tabular or image data.

To answer these questions, our project will evaluate the contribution of each modality through ablation studies, systematically removing or isolating modalities to determine which provide the most predictive value. We also plan on leveraging pretrained models to enact early fusion. Some examples include VisualBERT [8] to jointly encode visual and textual information for richer multimodal representations, and Perceiver [4] to combine and map visual data, texts, and numeric data into a single latent space. Our models' performance, specifically Mean Absolute Error (MAE), will be compared to prior work results, which are used as a performance baseline for traditional tabular or image-only models. Our models' performance will be compared against this benchmark to quantify improvement. A potential risk is data sparsity, as not all listings will have every modality. We can mitigate this by creating modality-specific embeddings and using masking or selectively encoding available data during the final regression step.

3.2 Preference Grounding and Tradeoff-aware Retrieval from Natural Language Queries

We propose a two-stage multimodal retrieval and re-ranking system with two key novelties absent from prior work. First, a preference parser explicitly decomposes queries into hard constraints (minimum bedrooms) and soft weighted preferences ("quiet street"), estimating a query-specific tradeoff structure rather than treating all clauses uniformly as in LLM-based filter extraction or flat CLIP [12] embeddings. Second, retrieved results are accompanied by evidence-grounded justifications linking each property to modality-level support (verifying whether geospatial data confirms transit access), addressing the black-box opacity of SeoulHouse2Vec [5] and MM-Embed [9], and integrating the natural language reasoning absent from Roomsemble's [7] image-only retrieval.

3.2.1 Planned Execution

We will evaluate the system in a component-wise manner before end-to-end testing, so improvements in final ranking can be attributed to parsing, retrieval, re-ranking, or explanation quality rather than just to an opaque pipeline. We formulate the task as constraint-aware ranking from natural-language housing queries: given a query and a multimodal listing pool, the system must satisfy hard constraints and optimize soft preferences. To reduce leakage, we will report random splits and both geography-aware and time-aware splits.

We will first build and validate a preference parser that outputs hard constraints, soft preferences, exclusions, and query-specific tradeoff weights, with schema-constrained parsing, deterministic checks for numeric fields and units, and confidence scores. We will measure parser performance separately using field-level accuracy and targeted error analysis.

Stage 1 candidate generation will combine hard-constraint filtering with text-based retrieval and be tuned for high recall at candidate set size K. Stage 2 will re-rank candidates using multimodal signals while enforcing hard constraints explicitly and modeling soft preferences with learned tradeoff-sensitive scoring. We will compare pairwise versus list-wise ranking objectives.

Finally, we will run baselines and ablations (modality, parser, objective) and stress-test robustness under missing modalities, cross-modal conflict, and spatial-temporal shift, while auditing whether evidence-grounded justifications are consistent with ranking decisions.

For our dataset, we will construct a multimodal Greater Boston residential listing dataset by collecting listing text, structured attributes, photos/floor plans, and transaction metadata from public real-estate platforms, then align these with parcel/location identifiers. We will augment listings with geospatial context from public sources (transit accessibility, neighborhood amenities, and neighborhood features), to support preference grounding beyond listing text alone. For evaluation, we will create a query-relevance benchmark using annotated natural-language housing queries and graded relevance judgments.

3.3 Context-Aware Optimal Land Use Prediction

When cities or private developers decide what to build on a vacant parcel, they face a complex optimization over physical constraints, regulatory rules, and contextual fit. Yet current methods are fragmented and manual, relying on site visits, zoning lookups, and experience-based intuition that does not scale across thousands of underutilized parcels. We propose a system for context-aware optimal land use prediction at the individual parcel level. The novelty is the shift from feasibility to recommendation: rather than predicting what is legally permissible [13, 3] or describing existing conditions [15, 10], our system predicts what should be built by jointly modeling physical constraints, zoning regulations, and contextual signals such as views, pedestrian traffic, and neighborhood character. Unlike commercial tools [1] that assess financial viability for a pre-specified use, we fuse satellite imagery, street-level photography, parsed zoning text, and geospatial context to rank all candidate uses jointly, with attention-weighted natural language explanations enabling planners to audit each recommendation.

The system has three components: (1) Physical Constraint Detection from satellite imagery, FEMA flood maps, and elevation data, producing binary constraint flags and height limits; (2) Contextual Feature Extraction from street-level photos, MBTA transit, and Census demographics; (3) Optimal Use Ranking via a cross-attention transformer fusing CLIP [12] embeddings (satellite + street view), Sentence-BERT [14] zoning embeddings, and contextual features into a ranked list over 8 land use categories. Feasibility is enforced through constrained decoding before softmax.

3.3.1 Planned Execution

We plan to use 500 Boston parcels developed 2015–2025 (MassGIS, MassBuilds, BPDA permits), chosen for parcel diversity. Inputs per parcel: satellite imagery, Street View, BPDA zoning text, MBTA proximity, and Census demographics. Each parcel is labeled with its actual developed use as ground truth. For quantitative evaluation, we report Top-k Accuracy, per-category F1 across 8 land use classes, and Constraint Satisfaction Rate (target $>90\%$). For qualitative evaluation, we conduct an expert agreement study with 3-5 MIT urban planning students on a held-out parcel set. Our central hypothesis is that jointly modeling constraints, zoning, and contextual visual signals outperforms any single-modality approach; we compare against three baselines: (1) a zoning-only rule system, (2) a random forest on tabular attributes, and (3) CLIP [12] visual similarity without regulatory embeddings. A key risk is limited labeled data, mitigated via data augmentation. And because land use recommendations carry real equity implications, we will check whether predictions systematically differ across neighborhood income levels, and flag any disparities for further review.

4 Conclusion

This proposal argues that real estate is not merely a prediction problem, but a multimodal reasoning problem. Across valuation, search, and land use recommendation, the same technical question recurs: how should a model fuse heterogeneous signals, decide what to trust, and remain robust when modalities are missing, conflicting, or strategically curated? If successful, this work could improve decision quality and reduce information asymmetry in a domain with broad economic and social consequences. By treating reliability, interpretability, and equity as first-class requirements rather than afterthoughts, we seek not just better predictions, but also more accountable use of AI in the built environment.

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Appendix

A Group Work Breakdown

We plan to share research tasks among teammates by dividing responsibilities across the different modalities and stages of the project while maintaining close collaboration to ensure integration of results. By 3/2, we aim to finalize our project idea and begin the initial data collection and preprocessing. Each teammate will take ownership of a subset of modalities, ensuring that all data, such as videos, images, 3D representations, and graph-based features, are cleaned, aligned, and ready for modeling. By 3/16, preprocessing should be complete. We will discuss strategies for fusing the modalities and begin modeling, experimenting with different architectures, with each teammate taking the lead on specific model components or data types. Between 3/16 and 3/30, we will continue refining the models, iteratively improving performance and integrating feedback from each teammate. By 4/13, we will finalize the model, evaluate its current performance using both quantitative and qualitative metrics, and begin preliminary analysis. Finally, by 4/27, we will complete the analysis to identify trends and insights, generate visualizations, prepare a clear presentation, and fully document our findings as a group.

B Github Link

`https://github.com/sophieelin/multimodal_ai`